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Virtual AI assistants in the manufacturing industry – innovation diffusion perspective

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Abstract

The study provides empirical insights into the innovation diffusion of AI assistants within the manufacturing industry. This study presents an innovative proof-of-concept (PoC) solution for a virtual AI assistant supporting human-cobot assembly tasks in manufacturing. Large Language Models (LLMs) are utilized for delivering operator instructions and Vision-Language Models (VLMs) support safety monitoring. The innovation diffusion analysis of the PoC solution identified several benefits, including improved assembly efficiency, reduced errors and hesitation, and flexible, hands-free instruction via natural language. Promising future applications of AI assistants in various industrial operations may include e.g. training, product documentation, and the sharing of tacit knowledge. To accelerate the diffusion of such innovations, further exploration is needed into key issues such as AI tool reliability, cybersecurity, compatibility with existing operations, and alignment with user needs.

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1. Introduction

The ultimate goal of this research is to help companies leverage benefits and potential of Artificial Intelligence (AI). Industrial companies are eager to leverage AI for competitive advantage, with Large Language Models (LLMs) like ChatGPT being used across industries such as software development, finance, healthcare, education, and manufacturing [1-3]. The use of Large Multimodal Models (LMMs), which handle both text and images, further expands industrial applications [4]. One common use is through AI-enhanced virtual assistants, which can perform tasks based on voice or text commands, or proactively respond to environmental or contextual analysis [5].

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However, further insights are needed into the factors influencing the adoption of virtual AI assistants in industrial operations and how they could be utilized more widely in various operations. Both technical innovations and factors related to user and company characteristics play a role in the diffusion of innovations [6] – the rate and extent to which new innovations are adopted. Consequently, the study aims to provide empirical insights into the following question: *How can the diffusion of virtual AI assistants be enhanced in the manufacturing industry?*

A design science approach, involving ten interviews, was employed for this research. First, a developed virtual AI assistant proof-of-concept (PoC) solution for human-cobot assembly work was analysed with the interviewees to gain insights into factors influencing innovation diffusion. The PoC solution includes LLMs for providing operator instructions through natural language and speech, VLMs for safety equipment supervision, and a visual Unity-animated 3D character as the assistant. Second, potential future use cases of AI assistants in the manufacturing industry were identified.

This study contributes by presenting an innovative virtual AI assistant PoC solution for human-cobot assembly task in manufacturing. The innovation diffusion analysis of the developed PoC solution highlighted key benefits, including improved assembly work efficiency, reduced mistakes and hesitation, and flexible, hands-free instructions through natural language. The most promising future use cases of AI assistants in general involve complex, knowledge-intensive industrial tasks, where extensive documents and instructions are fragmented across various systems and formats. AI assistants could play a significant role in e.g. training, product documentation, and sharing tacit knowledge.

2. Theoretical background

2.1. Innovation diffusion

Originally, innovation was defined as “the act of creating new combinations of existing resources to produce new products, services, or processes” [7]. Later on, the original definition has been complemented with aspects of not merely creating new ideas or products, but that the innovations are actually implemented and generating value for organizations and economies [8]. Our study focuses on technological innovations related to AI assistants to be developed and used in the manufacturing industry. Therefore, Diffusion of Innovations theory [6], is used as the theoretical starting point for this study.

Diffusion of Innovations theory categorizes individuals based on their adoption speed, presents the stages of decision-making process, and identifies key attributes of innovations that influence adoption [6]. This study foremost concentrates on the factors that affect the rate and extent of innovation adoption:

1. Relative advantage: The degree to which an innovation is perceived as better than e.g. the product it replaces,
2. compatibility: The degree to which an innovation is consistent with the existing values, experiences, and needs of potential adopters,
3. complexity: The degree to which an innovation is perceived as difficult to understand or use,
4. trialability: The degree to which an innovation can be experimented with on a limited basis before full adoption, and
5. observability: The degree to which the results of an innovation are visible to others.

Various barriers for the adoption of innovations include e.g. resistance to change due to fear of the unknown, limited resources for early adopters, and cultural factors that favor tradition or social norms [6]. The adoption of technological innovations is thus not only related to the characteristics of the innovation itself, but also to the adopter and organizational characteristics. Both these aspects are included in this study.

2.2. Virtual AI assistants for industrial use

AI is increasingly integrated into our daily lives, both in our personal time and at work. While the development of AI has been ongoing since the 1960s, recent advancements have made AI solutions accessible to mass markets. AI technologies are diverse, encompassing areas such as machine learning, natural language processing, machine translation, speech recognition, and knowledge representation [9]. In addition to previous generations of digital

assistants or software applications, AI-powered digital assistants are conversational agents that rely on speech and text, enabling natural language processing. More advanced AI-based assistants can also incorporate computer vision to interpret visual inputs. These assistants not only process and represent domain knowledge but can also learn and adapt through experience [10].

Using tools like ChatGPT, Amazon Alexa, and chatbots has become familiar to many of us in our daily lives. Concurrently, application of AI tools across various industries has increased, with companies actively seeking new ways to leverage AI for a competitive edge. This requires a deeper understanding of both how companies can effectively leverage AI and which factors developers of AI solutions must consider to ensure the successful diffusion of innovation. The scope of our research focuses on industrial, AI-enhanced virtual assistants.

A virtual assistant can be defined as “a software agent that performs actions or services based on commands or questions, which can be activated by voice commands or text-based inputs or can operate proactively based on the analysis of the environment or context” [5]. Virtual assistants can be enhanced with AI and LLMs, such as ChatGPT, enabling their use across various industries. In IT business, they can be used in software development [1, 11]. Earlier research concerning finance, healthcare, education, and manufacturing has identified several benefits of virtual assistants, such as round-the-clock availability, multilingual capabilities, real-time responses, low maintenance costs, ease of replication, extensive knowledge, and the ability to provide services through both text and speech [3]. In the manufacturing industry, virtual assistants can assist the workers in increasing productivity and ensure better data-based decisions [12] as well as reducing number of mistakes and difficulties of carrying out complex operations [13]. Also eyes-free and hands-free interactions, mobile assistance, the potential for various user interface types, and enhanced speed have been identified as benefits of AI [14].

In addition to LLMs, virtual assistants can also leverage Vision Language Models (VLMs) and Large Multimodal Models (LMMs), which handle both images and text are designed to work with multiple modalities of data, which means they can perceive, reason, and act across different kinds of inputs, and analyze, generate, and reason across these different types of data [4]. The combination of LLMs and VLMs/LMMs can unlock even greater potential benefits and use cases in the manufacturing industry. The methods can be applied in settings where an operator is working alongside a collaborative robot (cobot). AI enables cobots to understand and interpret human instructions through natural language processing, while also enhancing safety in industrial workplaces by detecting anomalies through machine learning [15].

In line with the Industry 4.0 paradigm, virtual assistants can play a crucial role in processing production data and delivering contextualized, real-time information to workers in the production environment [15]. Furthermore, our research aligns with the Industry 5.0 vision [16], focusing on facilitating new types of interactions and communication between operators and machines, which can be achieved through AI [14, 17]. In our study, the operator works alongside a cobot and they both use AI in performing the assembly task together. Cobots are particularly suited for applications where human-robot collaboration enhances efficiency, such as in small batch production and customization, whereas traditional robots are best suited for large-scale, high-volume production [18].

2.3. Research scope and questions

The use and R&D of AI are currently experiencing significant growth. However, industrial companies need a deeper understanding of AI’s potential and the ways it can benefit them through various use cases. To gain a competitive advantage, it is crucial to better understand the factors that influence the diffusion of innovation. Therefore, the main Research Question RQ for this study is: *“How can the diffusion of virtual AI assistants be enhanced in the manufacturing industry?”*

As part of the project, a Proof of Concept (PoC) solution for a virtual AI assistant was developed, providing a practical, concrete example of a technical AI-based innovation in this study. The diffusion of innovation for these types of solutions is analyzed through the “lens” of innovation diffusion factors. The PoC solution helps to concretize the research topic, offering insights both specific to the PoC itself and more broadly regarding how virtual AI assistants could be more widely adopted in the manufacturing industry in the future. Consequently, RQ1 concentrates on the developed AI assistant PoC, whilst RQ2 takes a wider perspective on potential future AI assistant use cases:

- RQ1: *“How is the AI assistant PoC solution perceived in terms of the factors influencing innovation diffusion: benefits and relative advantage, compatibility, complexity, trialability, and observability?”*

- RQ2: “How could AI assistants be used in the manufacturing industry in the future? What potential use cases can be identified for AI assistants within this industry?”

3. Materials and methodology

3.1. Virtual AI assistant PoC solution for human-cobot assembly work

Our Proof-of-Concept (PoC) solution features an AI-powered tool for hybrid AI-human-cobot teams performing assembly tasks in manufacturing. The PoC solution was developed and demonstrated in a laboratory environment. The worker interacts with a virtual AI assistant, represented as a human-like 3D visualization, which provides real-time assembly instructions (Fig. 1). The cobot delivers the correct components to the worker, while a camera in the workstation monitor the use of safety equipment. Key technical modules and features of the PoC solution include:

- Text-based LLMs and visual VLMs: LLMs are utilized for operator instructions and VLMs are employed for safety monitoring. The inclusion of “memory” allows the AI to remember prior interactions and conversations, ensuring more personalized and context-aware guidance for the operator.
- Local AI model deployment: The solution deploys Qwen2.5 (LLM) and Qwen2.5-VL (VLM) models locally, ensuring sensitive data and intellectual property remain secure, free from external servers or cloud storage, and mitigating data leakage risks.
- Gaming PC infrastructure: The system relies on GeForce RTX 4090 GPU with 24Gb of VRAM for interaction and AI models.
- Computer vision-based object detection in the cobot: The operator gives oral text-based instructions, and the AI agent enhanced cobot will automatically provide the necessary components, eliminating the need for jigs.
- Safety monitoring: The AI agent is equipped with object and operator detection features to monitor the use of safety equipment. If anyone in the work area is missing a helmet or vest, the system notifies the operator.
- 3D visualization with Unity: the assembled gearbox, settings, text input (questions), conversation history, and AI assistant are visualized in 3D on a VTT Power Wall setup, providing a comprehensive view of the task and a near life-sized AI assistant interacting with the worker through movement, gestures, and speech.



Fig. 1. UI of the AI assistant PoC solution.

The AI server (Fig. 2) is written in Python using asyncio library to provide asynchronous request-response communication over the TCP protocol. It combines LLM, VLM, STT and TTS models to produce text, speech, actions, and current assembly step as an output. The different AI clients can connect to the AI server at the same time, including safety camera, cobot, and virtual assistant. The human operator directly or indirectly interacts with the AI clients while wearing safety gear (helmet and vest) and input devices (PTT microphone and helmet-mounted camera). The human operator and AI clients produce text, speech, images, and videos as the input for the AI server.

The AI server maintains and updates a connected client list, broadcasting its responses to all the clients. Each client communicates with the server directly, and the clients do not communicate with each other. There is a request and response queues, ensuring that incoming and outgoing messages are processed in order. The model-specific code is contained in separate modules and can be exchanged, enabling use of different models for each role while keeping the server logic unchanged.

The AI server and all its clients exchange long JSON-based Unicode string messages as byte streams. Since the messages can contain audio samples and base64-encoded images, they are transmitted in chunks of 100 bytes. At the end of each message an End-of-Transmission character (EOT) is present as well as a bunch of Null characters as padding if the last chunk of the message is shorter than 100 bytes. The messages can contain the following content types: Text, Audio, Image, ImageWithText, ImageWithAudio, Video, VideoWithText, VideoWithAudio. Based on the content type, the message on the server side is being processed by different combinations of AI models, with the output of the one model being used as input for another. Since the server and the clients use the same JSON message structure, they only use the part of the message relevant to them. Mentioned architectural and procedural safeguards were implemented to mitigate hallucinations. Utilization of separate model for visual inputs reduced the complexity and cross-modal confusion. Also, soft safeguards were included by instructing both models with task-specific system prompts.

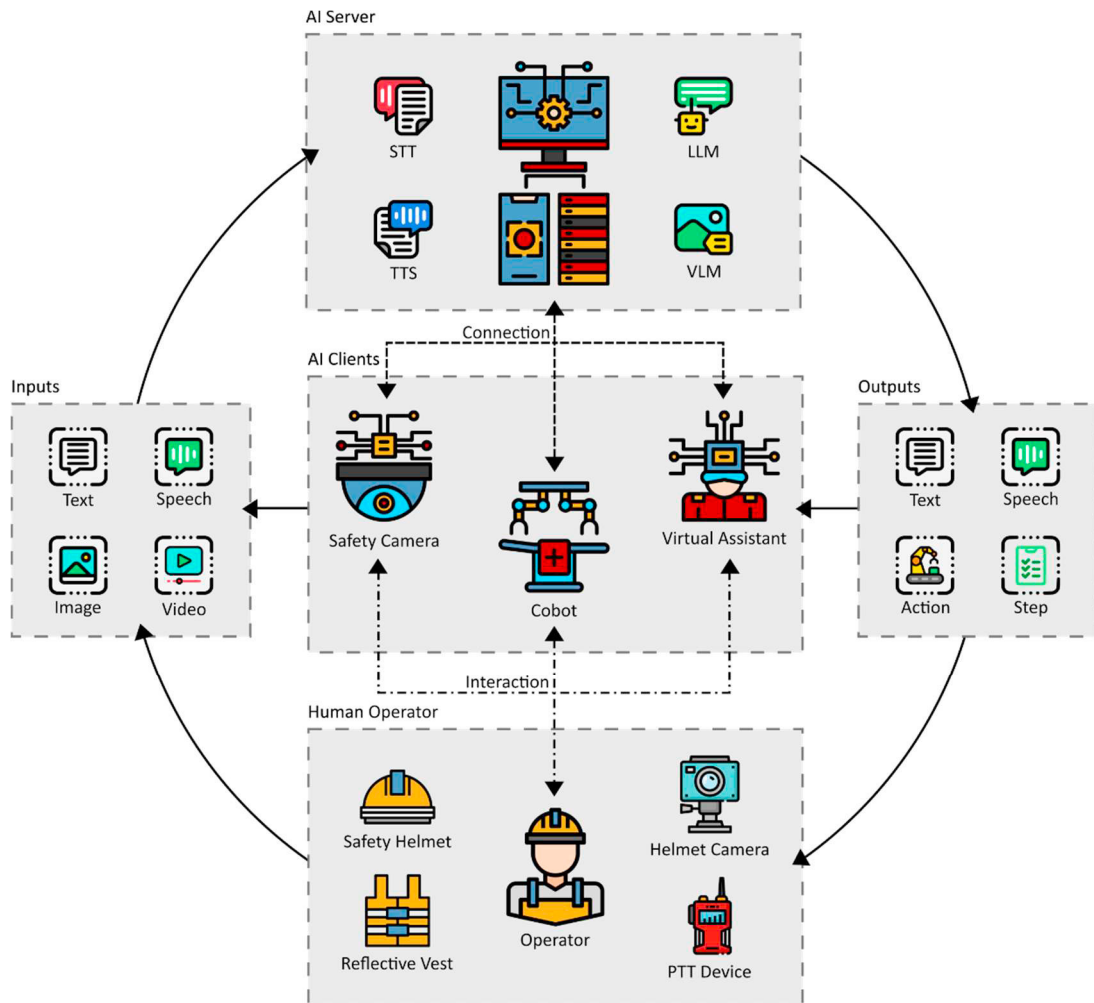


Fig. 2. System architecture diagram.

3.2. Methodological approach and study execution

This study uses the Design Science Research (DSR) approach, a pragmatic methodology widely applied in computer science, software engineering, and information systems [19, 20]. The DSR approach focuses on creating technology-driven solutions to significant business challenges [21]. The method was chosen because the study aims to develop an innovative AI assistant for industrial work and analyze the broader application potential of AI assistants in the manufacturing industry. “Artifacts” – such as software tools, models, or methods designed to address specific problems or needs within a given context [20–22] – are central to Design Science Research (DSR). In this study, the developed AI assistant represents the artifact. By developing the artifact and gathering feedback, the study explored the broader phenomenon of innovation diffusion, bridging the gap between theoretical research and real-world application [23] and using the collected perceptions to enhance future diffusion and R&D efforts.

This study was conducted as part of the large co-innovation project HiFive – “Meaningful Industrial Work in Hybrid Human-Technology-AI Teams” (2024–2026). The task reported in this paper was carried out between October 2024 and May 2025. The phases of this study were as follows:

1. PoC solution development: Virtual AI assistant for human-cobot assembly work,
2. presentation of the PoC solution to the informants with a PowerPoint presentation,
3. thematic in-depth interviews (n=10),
4. data analysis, and
5. reporting of results.

A total of 10 individuals were interviewed from six companies/organizations. The following table outlines the participating companies and organizations, interviewee profiles along with the number of interviewees:

Table 1. Study participant organizations and number of interviewees.

Company/ organization	Business/ operation field	Size	Interviewee profiles	n
Company A	Manufacturing	Large	R&D	2
Company B	Teleoperator	Large	Development director and manager	2
Company C	Manufacturing	SME	R&D	1
Company D	Engineering	SME	R&D	2
Company E	Innovation funding	Large	Research program lead	1
Company F	Research	Large	Research management, R&D management	2
In total				10

Participants provided informed consent and were briefed on the data privacy terms prior to the interviews. During the interviews, open-ended thematic questions were asked, focusing on factors affecting innovation diffusion: benefits, relative advantage, compatibility, complexity, trialability, observability, challenges, and further R&D needs. Ten interviews were conducted, each lasting between 30 minutes and an hour. The interviews were recorded, transcribed, and anonymized to ensure the confidentiality of the companies and interviewees in the reported findings. The interview data was analyzed using content analysis [24]. The data was categorized according to the main interview topics. Relevant quotes addressing the research questions were extracted and organized into a Word document table, condensing over 200 pages of transcripts into a 14-page summary while retaining key content [24]. Although thorough triangulation was not applied, data were collected from many companies, allowing versatile perspectives to emerge. Particular attention was given to views that were repeatedly expressed. In addition, innovative new ideas were collected, which some informants brought up. Key insights are presented in the Empirical Findings section, supported by illustrative quotations for transparency [25].

4. Empirical findings

The following presents the empirical findings based on the informants' perceptions of the developed AI assistant PoC solution and the identified potential future use cases for AI assistants. The findings are categorized according to the factors that influence innovation diffusion: benefits and relative advantages, compatibility, complexity, trialability, and observability.

4.1. Benefits and relative advantages

A key benefit highlighted by the interviewees regarding the presented AI assistant PoC solution was its ability to make assembly work more efficient and reduce errors: *"It gives the right components and tools, it decreases faults and hesitation of the assembler... It also notices if you are using a wrong tool or putting a part to a wrong place. It notices and helps even before the mistake has happened."* Avoiding mistakes and minimizing all sorts of waste also contribute to sustainability. Several informants shared the view that documents and instructions are often scattered across fragmented systems in various formats, making them difficult and time-consuming to locate. AI assistants can help address this issue: *"It increases efficiency when, instead of you going to look for the instructions somewhere in a strange place, them being in various formats, you can spar, discuss, and ask."* Also the flexibility in getting instructions from an AI assistant was appreciated: *"It is more flexible than rule-based instructions... you can jump [between phases] and you can ensure in the midst that you're proceeding in the right way."* The fact that the studied PoC solution was run on a local PC was seen as a major benefit, as it reduces the risk of confidential information being leaked.

Asking advice from the AI assistant can sometimes be more convenient than bothering a supervisor or colleague: *"It is still common for people to feel intimidated to ask questions, as they fear embarrassing themselves."* When discussing with AI assistant they can ask as many times as needed to understand. In addition to production, this can be useful also in internal logistics operations, e.g. in warehouse picking and route optimization. Another industry representative noted that an AI assistant can save supervisors' time by answering the same questions repeatedly asked by different workers. Unlike a supervisor or other colleague, who is not available 24/7, the AI assistant can provide support at any time, across all shifts. In this way, the AI assistant could also address frequently asked questions (FAQs), saving time for many.

Another advantage of AI assistants is their potential role in quality assurance. They could monitor and ensure that all necessary phases of the assembly process are completed correctly: *"Although it is quite standardized, there are employee-dependent deviations and their impact on end-product quality... which way is better."* AI assistants could gather and analyse data on various working methods, offering suggestions for the most efficient practices. This would help transfer tacit knowledge about best practices, addressing a significant challenge as experienced workers approach retirement.

A central benefit of industrial AI assistants lies in their ability to manage the vast amounts of documentation existing in fragmented systems and formats within companies. As one interviewee envisioned, AI assistants could add value to product and production planning and design: *"It would make life much easier if you could ask the AI: We have this product with these dimensions and background information. Please find out if we have a suitable tool that can be set to this machine to manufacture this item."* AI assistants could collect data throughout the production process and automatically generate product documentation. They could also compile reports on production, including statistics, for the production manager. Expanding on this idea, one company representative suggested that AI assistants could even assist salespeople by drafting offers, potentially reducing the need for immediate involvement from technical experts.

An interesting idea raised in the discussions was creating a feedback loop from assembly work to product design, ensuring products are designed for efficient and smooth assembly. Similarly, AI analysis and feedback loops from product use, repair, and maintenance could be directed to product design, identifying tasks that consistently cause issues and delays, ultimately driving product improvements.

When asked about the benefits of the AI assistant in the form of a 3D visualization, respondents shared various, sometimes conflicting, views. Some saw the study's focus on people and how their work could be supported as a valuable starting point: *"AI assistant is a workmate with which you can chat... I wonder whether there could be ways*

this makes work more fun, interesting, pleasant.” On the other hand, some informants felt that a human-like form could be unsettling and make workers anxious, as if they were being “watched”. Many questioned the necessity of a 3D assistant visualization, given that AI applications can be accessed through other interfaces. However, the ability to interact with the assistant hands-free, without wearable equipment, was seen as a significant advantage over mobile devices or AR glasses. Additionally, the 3D display of tools, numbers, and data was considered a valuable feature.

Regardless of the UI form, natural speech-based interfaces were seen as the future direction. However, noisy factory environments could pose challenges for speech recognition. In the PoC solution, cameras are used not only for cobot part handling task but also to ensure safety equipment is being used. While few safety-related benefits were mentioned, one informant suggested that AI could predict potential risks and hazards in the factory and alert workers. The view evolved from the AI assistant serving an individual to supporting an entire team.

The cobot component of the PoC solution generated less interest than the AI assistant itself. However, when combining assets like in this PoC, defining the optimal division of labor between humans, cobots, and AI is crucial to ensure they complement each other in terms of speed, precision, flexibility, and information processing. High information processing power was seen as a key advantage for AI: *“Where AI excels is handling large, known tasks – it’s done the same thing thousands of times, while a human may do it for the first time.”* In the future, AI could enhance human-cobot collaboration, making the cobot more flexible. For instance, you could ask the cobot to hold a component or pick up objects. Speech-based AI would integrate with robot control, adjusting cobot assistance based on user needs. The use of AI in robot/cobot control is expected to grow significantly in the coming years.

4.2. Compatibility with existing needs, operations, and company culture

All informants agreed that the AI assistant PoC solution is best suited for assembly tasks that are complex and involve multiple phases. It would be compatible with industries requiring frequent changes, product variations, and customization, aligning with trends toward increased knowledge intensity and digitalization. As products and machines grow more complex, AI could become a valuable tool, especially when tasks require a combination of competencies beyond human capability: *“Many products and machines are complex, requiring various technologies there may come a point when a human cannot always know everything.”* In such cases, AI could serve as a “collective brain” and memory, combining knowledge and capabilities that a single human cannot possess.

A key need for new industrial AI assistants, as highlighted by the interviewees, is addressing the workforce shortage. AI can boost productivity, allowing humans to focus on tasks where their presence is essential. AI solutions can also help companies in onboarding and training and by supporting multiple languages. AI could also be used to simulate and practice tasks before being applied in actual production. The interviewees viewed the compatibility of the PoC solution as generally good. Its implementation would not require major changes to the production system or cell, but rather serve as an addition to the existing environment: *“It doesn’t require changing the entire process or how we manufacture or assemble; it integrates into the current system to support the work without radical changes.”*

Regarding compatibility with company culture, interviewees noted that attitudes toward adopting new technologies vary significantly across workplaces, with personal and generational differences influencing acceptance. Interestingly, one company representative challenged the common view that younger generations are more eager to adopt new digital solutions. They suggested that older workers might embrace speech-based AI assistants if the technology felt human-like, stating: *“If the technology were very humanlike, they wouldn’t perceive it as technology but could interact with it as they would with a person.”*

4.3. Complexity of the solution

Regarding the complexity of the PoC solution, interviewees found the AI assistant easy to understand and use. One informant noted: *“Speech-based AI is so easy to use that everyone can use it.”* Another felt that industrial employees would quickly grasp the solution. Additionally, the fact that implementing the solution wouldn’t require significant changes to the environment or physical structures further facilitates its adoption.

4.4. Trialability

Regarding trialability, informants suggested that testing might not be necessary if the solution is simple to implement: *“You just plug it in, it's standardized, and you can take it anywhere, in the case we assume that it works.”* However, there is still significant development needed, especially for solutions integrating multiple assets (e.g., cobots, AI, cameras), like the PoC solution created in this study. On the other hand, many AI applications are easy to implement without extensive prior testing.

4.5. Observability of benefits

Regarding observability, or how quickly the benefits of industrial AI assistants are noticed, one informant shared: *“The benefits were apparent immediately, much faster than the time I spent learning to use it.”* Another manufacturing representative provided an example: *“I created an automatic 'planner' to optimize pallet filling. It took me almost three hours, but AI completed the task in ten minutes. I realized it makes no sense to compete against something so much faster.”* This was seen as a positive, as AI can handle back-office tasks, allowing workers to focus on other tasks. However, many companies are still in the early stages and need more experience with various AI solutions across use cases to fully realize their benefits.

4.6. Summary of the empirical insights and potential future use cases

The following tables summarize the findings at two levels of the study: Table 2 presents the innovation diffusion factors related to the AI assistant PoC solution, while Table 3 provides a broader view of potential future use cases for AI assistants in the manufacturing industry.

Table 2. Innovation diffusion factors regarding the AI assistant PoC solution.

Innovation diffusion factor	Empirical insights
Benefits and relative advantages	
Work task efficiency and time savings	Reduces hesitation and delays, provides accurate instructions quickly, eliminates the need to search fragmented systems or formats, saves time for supervisors and colleagues by minimizing the need for advice, availability 24/7 offering continuous assistance, hands-free operation with a speech-based user interface (UI), accelerates onboarding and training processes
Decreasing mistakes, quality insurance	Ensures the use of the correct components and tools with instructions and 3D model visualizations, provides flexibility in obtaining precisely the needed information through inquiry and repetition if needed, enables task simulation and practice before applying AI in actual production
Safety and cyber security	Cameras with AI serve as an enhanced safety and warning system, an AI model running on a local PC helps manage cybersecurity risks
Compatibility	
Needs	Pressure to achieve higher productivity, manufacturing products with high complexity and customization, increased knowledge intensity and digitalization in production processes, workforce shortages and high employee turnover, diverse cultural and language backgrounds among employees, ongoing need for extensive onboarding and training
Operations	Relatively easy to implement, as no major changes are required to the production system or cell, does not require significant investment
Company culture	A futuristic concept, particularly with the virtual visualization of the AI assistant, variation across companies and workers in terms of their eagerness to adopt new technologies
Trialability	Relatively easy to implement without extensive testing beforehand. However, developing a plug-and-play solution for the PoC that integrates multiple assets will require further R&D
Observability of benefits	Reduces the inconvenience of seeking advice. While the efficiency benefits are clear, more experience is needed to fully assess the feasibility and practical benefits.

Table 3. Potential future use cases for AI assistants in the manufacturing industry.

Operation or task	Potential future use cases for AI assistants
Product design	AI-assisted task analysis with feedback loops and improvement recommendations for product design
Production planning	Providing initial data for optimal manufacturing machine settings
Production	Identifies mistakes for quick correction and recovery, ensuring all phases are completed, analyzes working methods offering suggestions for the best and efficient practices, facilitates the transfer of tacit knowledge, assists with frequently asked questions, enables the cobot to provide “hands-on” support, such as holding a part for the operator, automates product documentation.
Training	AI-assisted training and onboarding
Internal logistics	Warehouse picking, mobile machine routing
Safety	Cameras monitoring the workstations for issues such as trespassers, trash, or incorrect items on the assembly table
Sales	AI-assisted drafting of the initial offer, followed by finalization with technical experts
Repair and maintenance	Instructions, AI-driven task analysis with feedback loops and improvement suggestions for product design.

5. Discussion and conclusions

The study makes three main contributions to the literature on Industry 5.0 [16] and virtual AI assistants [2, 3, 12–14]. As the first contribution of this study, we present the newly developed virtual AI assistant PoC solution, designed for use in assembly work in the manufacturing industry. This showcases a practical example of a hybrid AI-human-cobot team utilizing LLMs and VLMs for assembly task instructions and safety equipment monitoring. The virtual AI assistant, combined with 3D visualizations of tools and components and a speech-based interface operating in natural language, contributes to the realization of the Industry 5.0 vision [16]. The focus is on human and facilitating work within the manufacturing industry. The study introduced a novel form of hands-free, natural language-based interaction and communication between the operator and the cobot, utilizing AI [14, 17]. As a limitation, the study participants formed their views based on the concept description, and no user tests were conducted. Future research should include testing and implementing the PoC solution in real industrial environments to gain deeper insights into its feasibility, compatibility, and practical benefits.

Secondly, the study provides valuable empirical insights into the domain of virtual AI assistants and their industrial applications [2, 12–14] by applying the innovation diffusion perspective [6] to the study of virtual AI assistants in the manufacturing industry. The diffusion of AI solutions is, and will continue to be, highly relevant, as the adoption of AI is expected to accelerate across all industries in the coming years. Through the lens of innovation diffusion, rich empirical insights were gained, particularly regarding the benefits and compatibility with the needs of manufacturing companies, their production systems, and organizational cultures. The key benefits of the developed virtual AI assistant PoC solution include improved assembly work efficiency, reduced mistakes and delays, and the provision of flexible, hands-free instructions through natural language. As in this study, where an AI assistant for a single operator was explored, future research could focus on a wider team of people and AI assistants, all working as a hybrid team with complementary skills and strengths.

Compared to alternative technical solutions for virtual assistants, such as AR glasses or mobile devices, the studied virtual assistant was considered easier and more convenient to use due to its speech-based UI and the fact that no wearable equipment or mobile devices were required. This allowed users to keep their hands free for assembly tasks. Searching for instructions across fragmented systems, formats, or mobile devices would cause delays, which is a major advantage tackled by the studied PoC solution. However, The visualization of the virtual assistant was perceived as not providing significant additional benefits. While the 3D human-like character may have seemed unsettling to some, visualizing the tool and components to be assembled in 3D was regarded as a beneficial feature. Additionally, the cobot’s role in the modular solution was not regarded as as interesting as the AI assistant itself, although the potential of AI-based cobot control was recognized as a major area for future R&D.

As a third contribution, the study identified promising future use cases for AI assistants in complex, knowledge-intensive industrial tasks, where extensive documents and instructions are fragmented across various systems and formats. AI assistants could play a significant role in areas such as training, production planning, product documentation, and sharing tacit knowledge. Significant time savings were already reported, for example, in production planning-related calculations. In addition, using AI to analyze assembly task performance and maintenance tasks could help capture and share tacit knowledge and best practices. Even feedback for product design could be generated. However, these would require not just AI assistants performing tasks upon request, but AI agents [26] working more autonomously for data analysis and proactive operations. While adopting new technological solutions for efficient knowledge exchange, organizations can also consider ways to foster opportunities, openness, and trust among employees to enhance the flow of knowledge. Since this study primarily focused on AI assistants, we encourage further exploration and technological advancement of AI agents within the manufacturing industry. As an overall limitation of the empirical study, only ten interviews were conducted. Gathering more insights from industrial actors would undoubtedly help identify additional future use cases for virtual AI assistants. Furthermore, user testing can help validate the efficiency of different solutions and identify their benefits.

All in all, AI assistants were seen as beneficial and compatible with many current trends within the industry: need to increase productivity, lack of workforce, multi-language workplaces, constant need for training and onboarding, and passing tacit knowledge. To further enhance innovation diffusion of AI assistants in the manufacturing industry, challenges related to e.g. cyber security, reliability of AI models, and keeping up-to-date the documentation needed as basis for AI models, need to be carefully addressed. To foster widespread innovation diffusion, more emerging solutions tailored to the manufacturing sector are needed.

Companies need to overcome practical barriers related to hardware and software investments, as well as the necessary new competence building, and user training. Additionally, testing, validation, and “localization” in terms of language, business sector, and organizational culture must be carried out. Companies are advised to begin by identifying the most beneficial use cases for AI agents, then develop a roadmap, and experiment with proof-of-concepts to validate feasibility and benefits within their own operations. Through risk analysis and failure mode assessments, companies can mitigate the risks associated with the application of AI. Particular attention should be given to the potential leakage of confidential data, safety-critical tasks, and mitigating LLM hallucinations.

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